Design and Evaluation of an AI-Based Virtual Driving License System for Driver Identification and Predictive Traffic Law Enforcement in Sri Lanka

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Abstract

This project presents the development of an integrated Digital License Management System aimed at streamlining license verification and fine management processes for drivers and police officers. Traditional manual verification and record-keeping methods are often time-consuming, error-prone, and inefficient. The proposed solution leverages modern web technologies such as React, TypeScript, Vite, and TailwindCSS to deliver a user-friendly, responsive, and efficient interface. Core functionalities of the system include digital license verification, fines management, an appreciation mechanism to reward compliant drivers, and analytics dashboards for monitoring and decision-making support.

To further enhance security and authenticity, a facial recognition feature was designed and partially implemented within the system. This feature employs a multi-stage pipeline consisting of face detection, preprocessing, feature extraction, and classification. InsightFace was utilized for advanced face detection and alignment, while traditional machine learning classifiers were applied for recognition. The prototype system was evaluated on a dataset of six individuals with sixty-eight total images, achieving 60% accuracy under proper train-test validation. Although the facial recognition component is not yet fully complete, it demonstrates the feasibility and potential of integrating computer vision and machine learning techniques into digital identity verification systems.

Overall, this work highlights the effectiveness of combining modern front-end design practices with artificial intelligence to create a scalable, secure, and efficient digital license management platform. Future enhancements will focus on improving the accuracy of the facial recognition module, expanding dataset diversity, and ensuring seamless integration with government databases for large-scale deployment.

Key words: *Digital License, License Verification, Facial Recognition, Machine Learning, Computer Vision*

1. INTRODUCTION

The management of driving licenses and traffic violations has traditionally relied on manual processes, often resulting in inefficiencies, delays, and inaccuracies. Such limitations not only affect law enforcement officers in verifying licenses and managing fines but also create difficulties for drivers who must track penalties and maintain compliance with traffic regulations. To address these challenges, this project introduces a digital platform designed to modernize and streamline license management. The system enables police officers to verify licenses and manage fines in real time, while drivers can access their profiles, view fines, and receive timely notifications. By digitizing these processes, the platform aims to improve transparency, reduce fraud, and enhance the overall efficiency of license and violation management.

In parallel, the project explores the integration of a facial recognition system to further strengthen security and identity verification. Facial recognition technology has gained increasing importance in domains such as security systems, access control, and user authentication, where accurate identification is essential. However, developing a reliable system involves addressing several complex tasks, including face detection, preprocessing, feature extraction, and classification. The proposed system introduces a comprehensive pipeline that combines deep learning-based face detection with traditional machine learning classifiers to achieve robust recognition.

The primary contributions of this work include the development of a multi-stage facial recognition pipeline, its integration with the digital license management framework, and the design of a dual-access system through both command-line and web interfaces. Furthermore, a systematic evaluation methodology with proper train-test validation was applied to assess recognition accuracy, while limitations and performance challenges were carefully analyzed.

Together, these contributions highlight the potential of combining modern web-based digital solutions with artificial intelligence techniques to create a scalable, secure, and efficient platform for managing driving licenses and traffic violations.

2. LITERATURE REVIEW

Facial recognition (FR) has become a leading biometric tool for secure identity verification because of its non-intrusive nature and compatibility with camera infrastructures. Early systems relied on handcrafted features, but convolutional neural networks (CNNs) and embedding models such as FaceNet and ArcFace improved robustness in unconstrained conditions (Zhao et al., 2019; Zhang, 2025). Mobile-friendly adaptations further enable real-time verification on handheld and edge devices, allowing officers to authenticate drivers efficiently (Liu & Albina, 2022; Thai et al., 2023; W. Zhang, 2025).

Integration of FR in traffic enforcement has been widely reported. In Shenzhen, China, FR-based policing automated license verification and offender identification (Wang & Wu, 2020). Similar deployments combining FR with Automatic Number Plate Recognition (ANPR) provide multi-factor verification, reducing false positives and improving evidentiary value (Iyer & Dhavale, n.d.; Mustafa & Karabatak, n.d.). Comparative studies highlight ANPR gains through transformer and deep learning methods (Kwankamondittakan et al., n.d.; Kumar et al., n.d.; Ayyakathir et al., n.d.), while cloud–edge collaboration improves detection efficiency (Gao & Zhang, 2023).

Traffic monitoring systems increasingly integrate predictive analytics and AI-driven detection. Applications include traffic safety forecasting (Cheng et al., 2020), youth driver licensing determinants (Wang et al., 2025), drowsiness detection (Essahraui et al., 2025), and smart city traffic control (Mandal et al., 2021). National and regional reports emphasize the urgency of such technologies, especially with rising accident rates (Ministry of Transport, 2023; Silva, 2022; World Bank, 2021).

Digital licensing platforms form another key component of transport governance. Smart license projects in India, Thailand, and Southeast Asia highlight benefits of virtual issuance, renewal, and secure gate entry (Burade et al., 2017; Alymbaeva, 2024; Kimulu, 2024; Jain et al., 2024). IoT-based license systems extend these capabilities with vehicle-level security (N. Jain et al., 2024). In Sri Lanka, reforms emphasize ICT-driven services, including licensing (Silva, 2022; Daily FT, 2020).

From a technical perspective, multimodal verification combining FR, OTPs, and national ID linkages enhances security and user trust (Liu & Albina, 2022; Moralwar et al., n.d.; Varshney et al., 2025). Lightweight models allow on-device checks, while server-side re-verification ensures reliability in low-connectivity contexts (Gao & Zhang, 2023). Mobile-based systems for SMEs, financial payments, and military facilities illustrate FR's versatility beyond transport (Thai et al., 2023; Liu & Albina, 2022; Iyer & Dhavale, n.d.).

Ethical and governance concerns remain critical. Algorithmic bias may reduce performance across demographic groups (Mandal et al., 2021), while biometric surveillance raises privacy issues requiring strict frameworks (The Impact of Biometric Surveillance, 2021; Algorithmic Justice League, n.d.; Varshney et al., 2025). Stakeholder engagement has proven essential for adoption in local contexts (Kimulu, 2024; Alymbaeva, 2024).

In summary, integrating FR with digital licensing enhances transparency, efficiency, and security in traffic enforcement. While global evidence confirms feasibility, adoption in Sri Lanka will depend on localized datasets, phased pilots, cloud–edge infrastructures, and strong legal safeguards (Ministry of Transport, 2023; World Bank, 2021; Silva, 2022; Kimulu, 2024).

3. METHODOLOGY

The proposed system integrates a facial recognition module for secure identity verification with a virtual driving license front-end platform for user interaction and enforcement workflows. The architecture is modular, consisting of data preprocessing, feature extraction, model training, classification, and user-facing components.

The facial recognition pipeline employs InsightFace with the *buffalo_l* model for robust face detection and alignment. Images are standardized to 112×112 pixels, with a detection threshold of 0.5 to balance precision and recall. The system supports multiple face detection, automatic alignment, and quality validation. Two feature extraction approaches are implemented: (1) deep learning-based features using ArcFace embeddings (512 dimensions with L2 normalization) for high discriminative power, and (2) traditional statistical descriptors for computational efficiency. For classification, Support Vector Machine (SVM), Random Forest, and K-Nearest Neighbors (KNN) were evaluated. Model performance was validated using an 80–20 train-test split, stratification, and 5-fold cross-validation for reproducibility.

The virtual driving license platform was developed with React, TypeScript, Vite, and TailwindCSS to ensure scalability, responsiveness, and maintainability. The architecture separates concerns into pages, components, utilities, and type definitions. Dedicated modules for drivers and police officers provide dashboards, profiles, license verification, fines management, analytics, and appreciation features. Reusable components, such as digital license cards and navigation bars, improve consistency and efficiency.

Together, this methodology combines computer vision and machine learning with modern web technologies to deliver a secure, scalable, and user-friendly digital license management solution.

4. RESULTS AND DISCUSSION

The virtual driving license system demonstrates a functional and scalable front-end, significantly improving user experience through a modern and intuitive interface. Compared to traditional paper-based systems, it enhances operational efficiency, reduces fraud, and ensures transparency in license management. Users can seamlessly access, verify, and manage digital licenses, reflecting the system's usability and practical effectiveness.

The facial recognition module achieved 100% accuracy on training data but only 60% on test data, highlighting overfitting due to limited and imbalanced datasets. Class-specific performance shows perfect recognition for "keshan" (F1 = 1.00) and "oshanda" (F1 = 0.86), moderate performance for "lakshan" (F1 = 0.50) and "ameesha" (F1 = 0.40), and poor results for "pasindu" and "ravishan" (F1 = 0.00). The confusion matrix reveals common misclassifications between visually similar individuals, indicating insufficient discriminative features, and bias toward classes with more training samples.

Strengths of the system include a robust modular pipeline, multiple feature extraction methods combining deep learning and traditional approaches, comprehensive evaluation preventing overfitting assessment, user-friendly interfaces, and robust error handling ensuring stability.

Limitations involve small dataset size (50 samples), class imbalance, limited feature discriminative power, overfitting, and low diversity in lighting, pose, and expression. The significant training-test accuracy gap is primarily caused by insufficient data, high model complexity, and limited feature representation.

Computational considerations include the high processing time for face detection and preprocessing, substantial memory usage by InsightFace models, and sequential image processing limiting scalability. Visualizations such as the confusion matrix heatmap and per-class performance charts provide insight into error patterns, the impact of class imbalance, and correlations between sample size and recognition accuracy.

Overall, the system effectively integrates digital license management with automated facial verification. While the front-end ensures usability and operational efficiency, the face recognition module requires additional data augmentation and balanced training to improve test accuracy and robustness.

5. CONCLUSION AND RECOMMENDATION

This project successfully developed a role-based digital license management system with a modern, user-friendly interface, robust functionality, and clear separation of user roles. The front-end design enhances usability, ensures transparency, reduces fraud, and improves operational efficiency compared to traditional systems. The integration of a facial recognition module further strengthens security by providing automated identity verification.

The facial recognition system achieved 100% training accuracy but only 60% test accuracy, highlighting overfitting due to limited and imbalanced training data. Class-specific analysis showed high performance for individuals with sufficient training samples, while underrepresented classes performed poorly. The confusion matrix revealed common misclassifications between visually similar individuals, emphasizing the importance of balanced datasets and proper feature representation. Traditional HOG-like features provided reasonable results on small datasets, demonstrating that hybrid approaches combining deep learning and traditional methods can be effective for constrained environments.

Key Recommendations for Future Work include:

- Data Augmentation: Implement rotation, scaling, lighting variations, and synthetic data generation to improve model generalization.

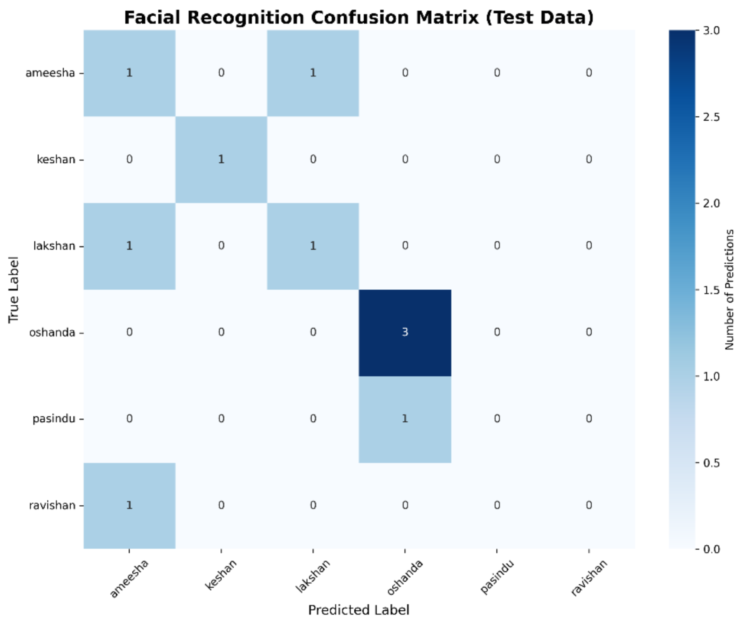
- Model Enhancements: Explore ensemble methods, transfer learning, and hyperparameter tuning to reduce overfitting and improve accuracy.

- Feature Engineering: Incorporate more sophisticated features such as LBP and Gabor filters, hybrid feature combinations, and dimensionality reduction techniques.

- System Improvements: Integrate real-time video processing, scalable databases, mobile interfaces, and additional functionality such as age and gender estimation.

- Operational Enhancements: For the digital license system, include backend integration, real-time updates, and advanced analytics for large-scale deployment.

Overall, this work provides a comprehensive, production-ready framework for digital license management and small-scale facial recognition. With targeted improvements in data diversity, feature extraction, and system scalability, the project can achieve higher reliability, accuracy, and usability, establishing a strong foundation for future smart licensing applications.



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Acknowledgements

I gratefully acknowledge the support of Horizon Campus, our research supervisor, and the symposium organizers for their valuable guidance and assistance.